

PAPER REVIEW • 20 JULY 2026

From Statistics to Reservoirs

Reservoir Computing, where it gets used, and how people actually clean ECG signals

9 papers scanned • 2 questions • 10 minutes



Three questions, ten minutes

01

Where RC came from

The road from ARIMA and ARMA to echo state networks — and why the jump happened.

02

How RC actually works

Three layers, one trained matrix, and the newer variants: NG-RC, minimal RC, physical RC.

03

ECG in practice

Noise sources, the four families of denoising methods, and which one won a head-to-head test.

How forecasting models evolved — and why RC appeared

1970s–90s

Statistical

AR • MA • ARMA
ARIMA • SARIMA

Linear, interpretable, cheap. Assumes stationarity.

2000s

Classical ML

SVR • Random Forest
GBM • Fuzzy TS

Handles nonlinearity, but needs manual feature engineering.

2010s

Deep Learning

LSTM • GRU • CNN
Transformers

Best accuracy, but black-box, data-hungry, expensive to train.

2001 →

Reservoir Computing

ESN • LSM
NG-RC • physical RC

Tiny training sets, linear optimisation, minutes not hours.

The survey's verdict: deep learning with modern optimisers improved accuracy by up to ~14% over classical methods across 150+ studies — but no family wins everywhere. Choose by data size, nonlinearity, and how much interpretability you need.

Reservoir computing in one picture

Throw a stone into a pond.

The ripples carry a record of when and where it landed. Throw a second stone and the patterns mix — the surface now encodes the whole recent history.

RC does exactly this: push your signal into a big, fixed, nonlinear dynamical system, let it echo, then read the answer off the surface with a simple linear fit.

TWO INDEPENDENT ORIGINS

- Jaeger (2001) — Echo State Network, from machine learning
- Maass et al. (2002) — Liquid State Machine, from neuroscience
- Unified as “reservoir computing” in 2005–07



Three equations — that's the whole algorithm

1 RESERVOIR UPDATE

$$\mathbf{r}(i+1) = (1-\gamma) \cdot \mathbf{r}(i) + \gamma \cdot \mathbf{f}(\mathbf{A} \cdot \mathbf{r}(i) + \mathbf{W} \cdot \mathbf{X}(i) + \mathbf{b})$$

The state mixes its own past with the new input. $\mathbf{f}$ is usually $\tanh$, γ is the leak rate. $\mathbf{A}$ and $\mathbf{W}$ are random and never change.

2 READOUT

$$\mathbf{Y}(i+1) = \mathbf{W}_{\text{out}} \cdot \mathbf{O}_{\text{total}}(i+1)$$

The output is a linear combination of the reservoir state — or, in NG-RC, of features built straight from the data.

3 TRAINING — TIKHONOV / RIDGE

$$\mathbf{W}_{\text{out}} = \mathbf{Yd} \cdot \mathbf{O}^T \cdot (\mathbf{O} \cdot \mathbf{O}^T + \alpha \mathbf{I})^{-1}$$

One closed-form matrix inversion. No gradients, no epochs, no BPTT. α is the ridge parameter — it prevents overfitting and buys noise tolerance.

WHY THIS MATTERS

~400

training points were enough for a Lorenz-63 forecast

1

matrix inversion replaces the entire training loop

0

vanishing-gradient problems — there is no gradient

What makes a reservoir a good reservoir



Reproducibility

Similar inputs must give similar outputs. This is the reservoir's ability to generalise beyond the training data.



Separation

Different inputs must give different outputs. A reservoir that always answers the same thing is reproducible — and useless.



Fading memory (echo state property)

Recent inputs matter most, old ones fade, and the influence of the initial state must vanish. The memory timescale should match the signal's.

A CAVEAT WORTH REPEATING TOMORROW

The folklore says a reservoir should sit “at the edge of chaos”, near an instability where the dynamics are richest. There is a real intuition behind it: ordered regimes erase the input too fast, chaotic regimes destroy separability. But counterexamples exist, and Jaeger himself calls the edge-of-chaos rule a myth — neither well-defined mathematically nor confirmed empirically. Treat spectral radius as a hyperparameter to tune, not a law to obey.

RC vs. the usual suspects

LSTM / GRU

Gated cells, trained end-to-end

- All weights learned by BPTT
- Needs large datasets
- Slow to train, hard to tune
- Strong on long, rich sequences

Transformers

Self-attention over the whole window

- Best raw accuracy at scale
- Very data- and compute-hungry
- Weak interpretability
- Overkill for short univariate signals

Reservoir Computing

Fixed dynamics + linear readout

- Only W_{out} is trained
- Works on tiny training sets
- Trains in seconds; runs on hardware
- Excellent on chaotic / dynamical data

Rule of thumb: if the data comes from a dynamical system and you have little of it — try RC first.

NG-RC: the reservoir that isn't there

A linear reservoir with a nonlinear readout is provably just as powerful as the classic nonlinear reservoir — and it is mathematically identical to a nonlinear vector autoregression (NVAR). So the network can be deleted entirely, and the feature vector built straight from the data:

c

a constant term

+

O_{lin}k time-delayed observations,
spaced by s

+

O_{nonlin}polynomial (quadratic, cubic...)
combinations of those

WHAT YOU GAIN

- No random matrices → reproducible and interpretable
- Far fewer metaparameters to search over
- W_{out} can be read like an equation of the dynamics

REPORTED GAINS

33–162×

cheaper than an already-efficient traditional RC (Lorenz-63)

> 10⁶×

cheaper than a high-accuracy traditional RC

45

feature components were enough to infer an unseen variable

When you can only measure one signal

Real sensors rarely give you the full state vector. What if only a single scalar time series is observable?

1 Reconstruct the state

Takens-style time-delay embedding. Dimension from false nearest neighbours, delay from average mutual information.

2 Raise the nonlinearity

In delay coordinates, low-degree monomials fail. NG-RC only works once high-degree monomials are included.

3 Swap the basis

Replace monomials with Chebyshev polynomials of the first kind → “NG-RC-Ch”. An orthogonal basis behaves far better numerically.

+2

Lyapunov times of extra valid prediction, at the same degree of nonlinearity, purely from switching monomials → Chebyshev polynomials. Tested on Rössler, Lorenz, and a real electronic chaotic oscillator. The models also reproduced long-term behaviour — bifurcation diagrams and return maps, not just short traces.

Killing the randomness

Random reservoirs are cheap to build but hard to analyse and impossible to reproduce exactly. Two papers attack this from different directions.

Minimal RC

Ma, Prosperino & R  th — Sci. Reports 13:12970 (2023)

- Feed each coordinate combination into the reservoir separately
 - Replace the random matrix with a block-diagonal matrix of ones — it acts like average pooling
 - Move the nonlinearity to the readout: append higher orders of the state before ridge regression
- ≈ 7.0 Lyapunov times of forecast horizon on Lorenz — about 2.5× the average of traditional RC, NG-RC and SINDy, with long-term errors ~9× and ~40× smaller.**

Fully deterministic RC

Viehweg, Poll & M  der — Sci. Reports 15:17695 (2025)

- Replace the random input mapping with a parameterised Logistic map (TCRC-LM) or Chebyshev map (TCRC-CM)
 - New activation: the Lobachevsky function — periodic and non-saturating, unlike tanh
 - One deterministic run replaces 15–100 random seeds
- Beats ESNs by up to 99.99% on non-chaotic and 87.13% on chaotic series (Mackey-Glass), and cuts TCRC-ELM error by up to 68.08% — at ~8× the runtime of a single ESN run.**

WHERE IT GETS INTERESTING

The reservoir doesn't have to be code

Any nonlinear dynamical system with fading memory can serve as the reservoir. Training stays a linear regression on the outside — so the “network” can be physical hardware.

Electronic

Memristors — resistors with memory that mimic synaptic plasticity

Photonic

On-chip silicon photonics and delay-line lasers; million-word/s classification

Spintronic

Skyrmion and magnetic-texture dynamics

Quantum

Quantum reservoirs on near-term devices; weak and projective measurements

Soft matter & robotics

A soft body's own mechanics as computation — morphological computing

Biological

Cultured biological neurons acting as generalisation filters

Case in point: a reservoir built from a single discrete-memristor chaotic map hit 99.99% accuracy on nonlinear recognition and RMSE 0.0974 forecasting the Logistic map — with fewer samples and lower training cost than existing RC systems.

Deng et al., Chaos, Solitons & Fractals 194:116230 (2025) · Tanaka et al., Neural Networks 115 (2019)

Where RC is actually being used



Chaotic forecasting

Lorenz, Rössler, Mackey-Glass; attractor and Lyapunov reconstruction from data alone



Weather & climate

Spatiotemporal chaos, turbulent convection, tipping-point extrapolation



Wireless comms

Channel equalisation — the original Jaeger & Haas result



Digital twins

Model-free surrogates; predicting and preventing extreme events



Energy & environment

Wind speed, sunspots, solar output, temperature series



Signal & speech

Audio processing, waveform and handwritten-digit recognition



Finance

Exchange rates and market trajectories from short histories



Biomedical

ECG heartbeat classification and wearable monitoring

RC has already been pointed at ECG

ESN for ventricular beat classification

Single-lead ECG under the AAMI inter-patient protocol, designed explicitly for Medical-IoT wearables. Fast, scalable, real-time.

ESN on MIT-BIH, five classes

Raw unprocessed 1-D heartbeats. The best model exceeded 96% overall accuracy (90.2–99.1% per class) — above several CNN baselines, at a fraction of the training cost.

Cascaded NG-RC + conventional RC

A normalised ESN stabilises training by normalising reservoir states before the activation; evaluated on the MIT-BIH arrhythmia database.

WHY IT FITS

- ECG is a nonlinear, non-stationary time series — RC's home turf
- Training cost low enough for on-device, patient-adaptive models
- Robust to input noise by construction, thanks to ridge regularisation
- Maps cleanly onto memristor and photonic hardware for wearables

In every one of these, denoising happens before the reservoir.

The ECG — and everything that ruins it

The signal

Weak and non-stationary, roughly 0.05–100 Hz with most energy below 35 Hz.

P wave 0.67 – 5 Hz

T wave 1 – 7 Hz

QRS complex 10 – 50 Hz

Baseline drift 0 – 0.3 Hz

The noise

It overlaps the diagnostic bands — which is why you can't just low-pass everything.

- Baseline wander — breathing, movement, electrode drift
- Power-line interference — 50/60 Hz and its harmonics
- Muscle contraction (EMG) — broadband, millivolt-level
- Motion artifacts and electrode-contact noise
- AWGN — especially in wireless and wearable recording

Over-filtering is as dangerous as under-filtering: cut above 35 Hz and the trace looks beautiful — while ST-segment and QRS detail you need for diagnosis quietly disappears.

Four families of ECG denoising

A

Classical digital filters

High-pass (0.5 Hz, order 4) for baseline wander · notch at 50/60 Hz for PLI · Chebyshev Type II · moving average · Savitzky-Golay

Fast, linear-time, trivial to deploy. Fixed bands, so they can't adapt to non-stationary noise.

B

Model-based / state estimation

Kalman filter · Extended KF · Cubature KF · Adaptive Dual Augmented EKF (ADAEKF)

Adapts to time-varying noise in real time; CKF avoids Jacobians. Needs a dynamic model of the ECG.

C

Transform / decomposition

Stationary Wavelet Transform · Empirical Wavelet Transform · EMD and ICEEMDAN · S-transform · quantum smoothing filters

Multi-scale and adaptive. SWT skips downsampling → best feature preservation. EMD suffers mode mixing.

D

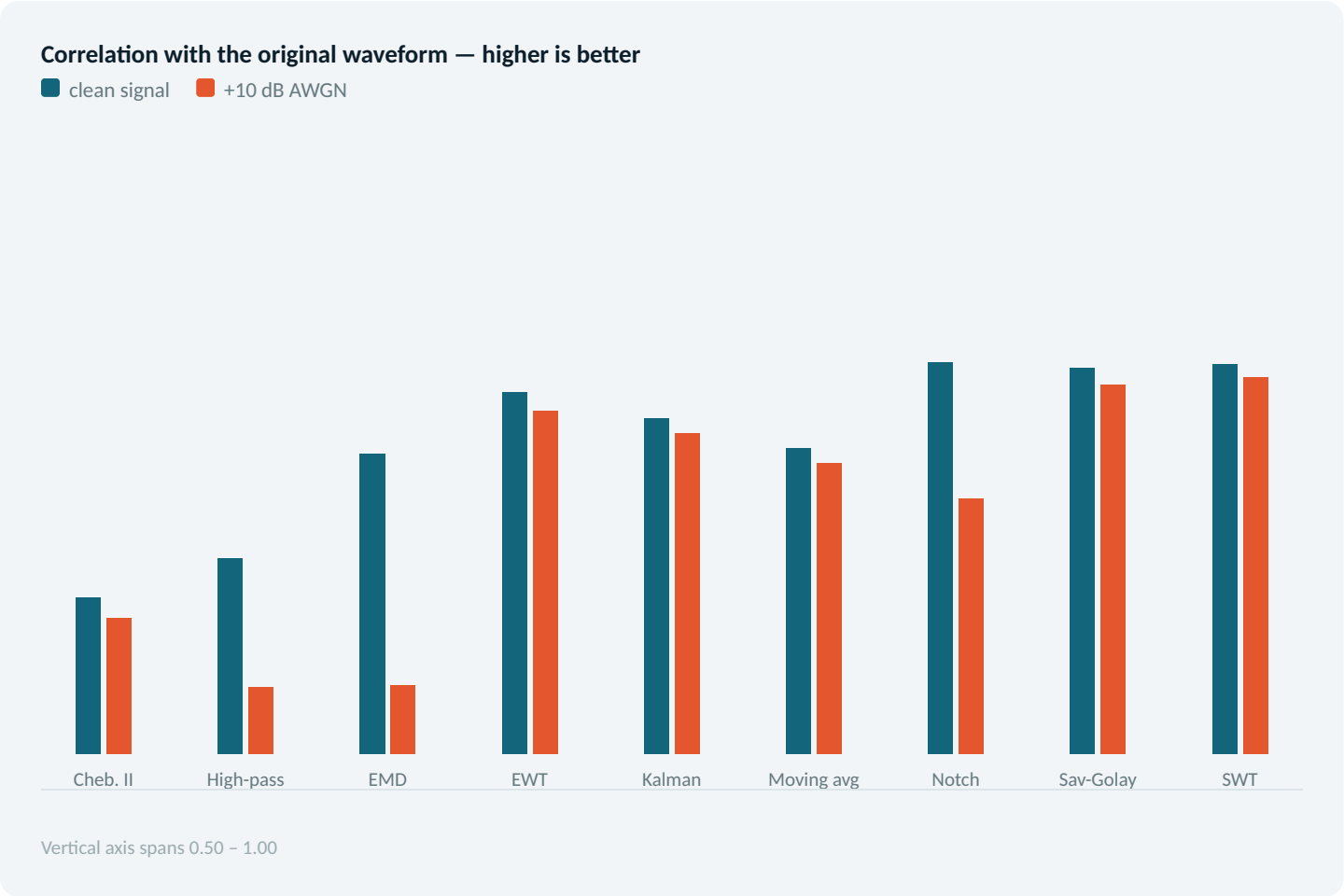
Learning-based

Denoising autoencoders (incl. fully-gated / self-ONN) · CNN-SWT hybrids · GANs · transformer encoders · non-local means

State-of-the-art quality and generalisation — but large, slow, resource-hungry, and needs clean references.

Hybrids dominate current practice: wavelet + median, ICEEMDAN + optimisation, wavelet + PSO tuning, CNN regularised by SWT.

One study, nine methods, 516 × 12 PTB records



READ IT THIS WAY

- Notch is near-perfect on clean data and collapses under broadband noise — it only knows one frequency
- EMD does the same: 0.88 → 0.59 once real noise arrives
- SWT holds up best: PRD 1.9–7.7% and correlation 0.98–0.998 across all conditions
- Savitzky-Golay is the surprise runner-up — and it is far cheaper

Cost vs. fidelity — and a pipeline we could try

Measured runtimes

Mean execution time per segment, Xeon E5-2670 v3

Notch	0.50 ms
Savitzky-Golay	0.70 ms
High-pass	0.91 ms
Chebyshev Type II	3.31 ms
SWT	intermediate
EWT, then EMD	slowest by far

Classical filters run in $\sim O(N \cdot d)$ with small order d . Adaptive methods buy fidelity with compute.

A proposed starting pipeline

- 1 Resample to a uniform rate — 500 Hz is a good balance
- 2 Notch at 50 Hz — cheap, and it kills the PLI outright
- 3 High-pass at 0.5 Hz, order 4, for baseline wander
- 4 SWT denoising (rbio3.9, level 5, threshold 0.5) as the workhorse
- 5 Score with SNR, PSNR, PRD, RMSE and correlation — not by eye
- 6 Only then feed the RC / ESN classifier

What I'd take out of this pile

1

RC is a training trick, not an architecture

Freeze the dynamics, train one linear layer. ESN, LSM, NG-RC and memristor RC are all variations on that one idea.

2

The field is deleting its own randomness

NG-RC removes the reservoir; minimal RC and TCRC-LM/CM remove the random matrices. Interpretability and reproducibility are the driver.

3

RC's edge is small data and cheap hardware

Hundreds of training points, seconds of training, and it maps onto physical substrates. That's the argument for wearables — not raw accuracy.

4

For ECG, preprocessing sets the ceiling

SWT was the most robust of nine methods, but the real lesson is that noise and signal share bands — so measure distortion, don't eyeball it.

Open question for tomorrow: can an NG-RC do the denoising itself, instead of sitting downstream of a wavelet filter?

Papers behind this deck

- te Vrugt, M. (2024). *An introduction to reservoir computing*. arXiv:2412.13212.
- Gauthier, D. J., Bollt, E., Griffith, A. & Barbosa, W. A. S. (2021). *Next generation reservoir computing*. Nature Communications 12, 5564.
- Ratas, I. & Pyragas, K. (2024). *Application of next-generation reservoir computing for predicting chaotic systems from partial observations*. Physical Review E 109, 064215.
- Ma, H., Prosperino, D. & R  th, C. (2023). *A novel approach to minimal reservoir computing*. Scientific Reports 13, 12970.
- Viehweg, J., Poll, C. & M  der, P. (2025). *Deterministic reservoir computing for chaotic time series prediction*. Scientific Reports 15, 17695.
- Deng, Y., Zhang, S., Yuan, F., Li, Y. & Wang, G. (2025). *Reservoir computing system using discrete memristor for chaotic temporal signal processing*. Chaos, Solitons & Fractals 194, 116230.
- Houssein, E. H., Mohamed, M., Younis, E. M. G. & Mohamed, W. M. (2025). *Artificial intelligence and classical statistical models for time series forecasting: a comprehensive review*. Journal of Big Data 12, 271.
-     m, N., Val'ko, D., Balogh, Z., Mado  , B. & Hurtuk, J. (2025). *Comparative evaluation of filtration techniques for ECG signal denoising with emphasis on stationary wavelet transform*. Scientific Reports 15, 42514.
- External sources: ESN-based ventricular heartbeat classification (2019) · ESN heartbeat classification on MIT-BIH, Computation 14(2):49 (2026) · Fully-Gated Denoising Autoencoder, Sensors 25(3):801 (2025) · ECGDnet transformer denoising, Signals 6(2):26 (2025).